Independent Retrieval Does Not Reduce False Endorsement on SciFact

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Abstract

False endorsement—predicting SUPPORT when the gold label is not support—is a costly error in scientific claim checking. We ask whether a staged pipeline that reconciles the original citations (D_0) with independently retrieved abstracts (D_1) reduces it relative to a same-evidence, evidence-first baseline (B2). Four isolated, fail-closed roles run as stateless calls to one model.

On 300 SciFact development claims scored with GPT-6 Luna under a frozen configuration, B2's false-endorsement rate is 0.011 and the validator (V) reaches 0.000. The reduction rests on 2 versus 0 discordant pairs (exact McNemar $p = 0.50$); the 95% interval (-0.012, [-0.031, 0.000]) includes zero. SUPPORT recall falls from 0.530 to 0.383 (66 missed supported claims against 19 missed contradictions). A 300-claim held-out run repeats the pattern, with 38 of 300 V requests failing closed. The independent arm alone (V-D1) endorses *more* falsely than B2 on that held-out sample (0.032 against 0.017). A second model (MiniMax M3.1 Flash) moves in the same direction (0.033 to 0.016; SUPPORT recall 0.855 to 0.692), again with an interval covering zero.

Independent retrieval finds no annotated document that was not already cited (unique yield 0.00). Most V errors come from disagreement between the two scopes, which the reconciler maps to underdetermined by construction. V is more conservative than B2, not more accurate, and this corpus cannot test whether independent retrieval repairs a wrong citation.

Artifact. Every table and figure is generated from saved prediction files by `scripts/paper/`. The corpus, splits, prompts, and configurations are in the accompanying repository (Appendix E). The primary development and held-out runs use the same frozen configuration; a later re-run of the development sample after a retry policy was added is reported as a second run, not as the primary result.

1 Introduction

A claim checker that reads only the citations offered with a claim inherits the generator's evidence. If those citations are wrong, or support a slightly different claim, the checker cannot notice. One proposed remedy is independent retrieval: search again, judge the new passages separately, and reconcile the two scopes. Recent work on search-augmented factuality evaluation [Wei et al., 2024], self-check agents [Li et al., 2026, Hirsch et al., 2026], and retrieval-augmented verifiers [Yue et al., 2024, Zheng et al., 2024] points in that direction. The assumption is rarely tested against a same-evidence control on scientific claims, where the original citations are often correct and where abstention can be the right answer.

We build a staged validator (**V**) that separates four roles—a sealed claim-blind reader, a claim-visible judge over four labels, a citation auditor over the original references, and a reconciler—across two evidence scopes: the original citations (D_0) and independently retrieved abstracts (D_1). The comparison is against a same-evidence baseline (**B2**) that reads a neutral question over the original citations in an isolated reader and then compares. B2 is the evidence-first control, not the strongest same-evidence configuration we ran: a claim-visible judge over the same citations (B1) has higher micro-F1.

The specified primary outcome is false endorsement: the rate at which a method predicts SUPPORT for a claim whose gold label is not support. The observed answer is inconclusive. On the frozen development sample the paired difference between V and B2 is -0.012 ([-0.031, 0.000]), which includes zero, and an exact McNemar test on the 2 versus 0 discordant pairs gives $p = 0.50$. The held-out sample repeats the pattern. V reaches zero false endorsements on both Luna samples by withholding support:

SUPPORT recall drops from 0.530 to 0.383 on development, and on the held-out sample 38 of 300 claims fail closed. Both false endorsement and support recall are needed to judge the trade-off.

Three observations frame the result.

- **This corpus cannot test the motivating mechanism.** In SciFact the annotated gold documents are a subset of the cited documents, so independent retrieval's unique yield is 0.00. The experiment measures what happens when a second reader of the same literature is added to gold citations, not whether search repairs a wrong citation.
- **The conservatism is a reconciliation policy, not a retrieval gain.** When the two four-way labels agree, V matches B2. When they differ on the scored three-way label, V abstains by construction and is right only as often as those claims are gold NEI. About half of the recorded "conflicts" are unaddressed-versus-underdetermined: both scopes already say NEI.
- **The cost of that policy is missed support.** On the full development sample V misses 66 supported claims and 19 contradicted ones. A thirty-claim subset of four closed-weight models from one vendor lineage concentrates the loss on REFUTE; that subset does not match the full-sample class breakdown and is not a replication.

The contribution is a reproducible measurement of what independent retrieval and evidence-first staging do, and do not do, on SciFact, together with the artifact that makes the measurement checkable. We report it as an inconclusive result with a mechanism, not as evidence that independent retrieval does not help in general.

2 Related Work

2.1 Scientific claim verification

FEVER [Thorne et al., 2018] established the claim–evidence format this paper follows; FEVEROUS [Aly et al., 2021] extended it to mixed evidence. SciFact [Wadden et al., 2020] pairs scientific claims with abstracts and rationale sentences and supplies support, refute, and not-enough-information labels. Official SciFact systems are retrieval-plus-rationale pipelines scored at the abstract and sentence level: VeriSci is the original cascade in that paper, and MultiVerS [Wadden et al., 2022b] is the usual supervised reference (about 72.5 abstract-level label-only F1). Those numbers are not interchangeable with the claim-level three-way micro-F1 we report. SciFact-Open [Wadden et al., 2022a] extends the task to open-domain retrieval; we use a frozen corpus only. Missci [Glockner et al., 2024] and RAGTruth [Niu et al., 2024] sit at neighbouring boundaries (fallacy reconstruction; context faithfulness), not as scored competitors.

LLM-only scientific verification is no longer empty. Alvarez et al. [Alvarez et al., 2024] show that GPT-4 few-shot prompting on citance–abstract pairs is close to MultiVerS under the official SciFact metrics. TripleCheck [González et al., 2025] reports 0.70 claim-level F1 on SciFact development in a zero-shot setting. SciClaims [Ortega and Gomez-Perez, 2025] is an end-to-end generative biomedical pipeline. We do not reproduce these systems; we cite them so that a prompt-level B2/V comparison is not mistaken for the only LLM result on SciFact. Vladika and Matthes [Vladika and Matthes, 2024] compare knowledge sources for open-domain scientific verification and motivate asking whether a different evidence source changes verdicts.

2.2 LLM verifiers, retrieval, and citation

Chain-of-Verification [Dhuliawala et al., 2024] isolates verification questions from a draft answer; B2 deliberately imitates that isolation over the original citations. SAFE [Wei et al., 2024] verifies atomic facts with search and is the reference point for the independent-retrieval arm. The primitive the judge implements—is this statement supported by these documents?—is a standard LLM-as-a-judge operation [Zheng et al., 2023]. FActScore [Min et al., 2023] and MiniCheck [Tang et al., 2024] score generated text against grounding documents; we check an already-atomic scientific claim against retrieved abstracts, so

we inherit the isolation idea and not the metric. Self-checking agents such as MARCH [Li et al., 2026, Qwen-Applications, 2026] add information separation between proposer and checker. Blame-localization work [Hirsch et al., 2026] attributes citation errors in a trace; we evaluate final conclusions and cite that work as the reason not to claim trace-level credit. RAFTS [Yue et al., 2024] is a retrieval-then-judge verifier; RAV [Zheng et al., 2024] is a trained FEVER pipeline whose thesis is that retrieval quality dominates, which is closer to our negative retrieval finding than to our staging story. Citation adequacy is its own evaluation problem [Gao et al., 2023]; our auditor rates each D_0 citation but we do not report an ALCE-style citation-precision number.

When two evidence scopes disagree, the model is in a knowledge-conflict regime. Xie et al. [Xie et al., 2024] show that models follow coherent external evidence but revert to a parametric prior when some of the context agrees with that prior; ClashEval [Wu et al., 2024] quantifies the same tug-of-war. The reconciler's default to underdetermined is a conservative policy in that literature, not a new theory of conflict.

2.3 Abstention and selective use

The precision–coverage curve we plot is the risk–coverage trade-off of selective classification [El-Yaniv and Wiener, 2010, Geifman and El-Yaniv, 2017]: a reject option that trades coverage for lower error among accepted claims. Feng et al. [Feng et al., 2024] study when an LLM should abstain because of a knowledge gap; Yang [Yang, 2026] is a concurrent fact-checking application. Neither paper is the formalism. Calibration theory [Guo et al., 2017] distinguishes verbalised confidence from a probability fitted on held-out data; we fit a temperature only for the one condition that verbalises confidence (B1) and do not treat that as a result for V. We keep a scientific "underdetermined" verdict and a system-level "defer" action in separate fields.

3 Problem Formulation

3.1 Inputs and evidence scopes

Let c be an atomic scientific claim. A claim carries a source span, a normalised statement, and a *neutral question*: a single interrogative that preserves population, comparator, units, and causal scope but does not contain the asserted answer. Two evidence scopes are available:

- D_0 — the documents the original report cited. A judgment over D_0 answers: *is the original citation set faithful to the claim?*
- D_1 — documents retrieved independently from a frozen corpus using the neutral question, with the asserted answer, the report, and D_0 hidden. A judgment over D_1 answers: *what does independent retrieval establish?*

A verdict is always relative to a stated scope, corpus, and search budget. In SciFact the "original citations" are the benchmark's cited documents, which contain the annotated gold evidence. That is a property of the test bed, not of a generator that might have cited the wrong paper.

3.2 Four labels

Internally the system uses four labels, defined relative to a specified evidence collection:

Supported
Relevant evidence establishes the scoped assertion and no unresolved material counterevidence remains in the reviewed pool.

Contradicted
Relevant evidence establishes an incompatible assertion.

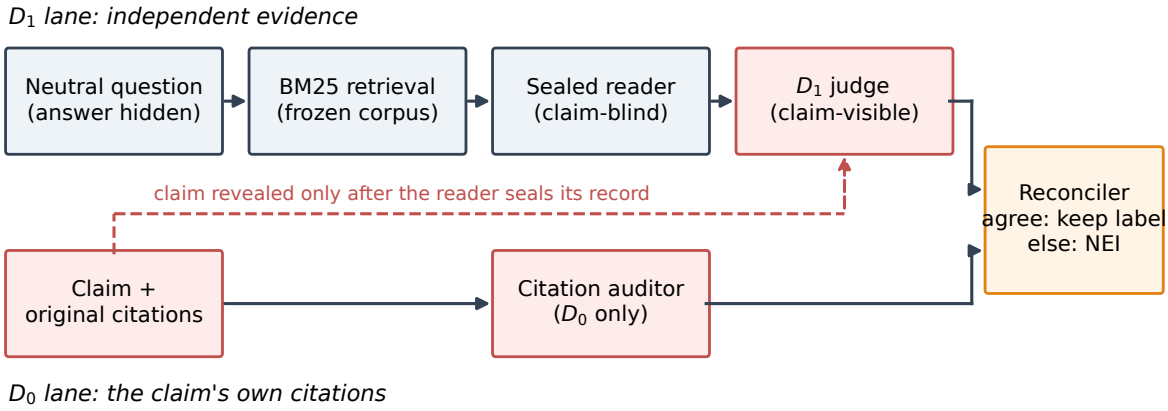


Figure 1: The staged validator. The D_1 lane retrieves and reads from the neutral question; the claim enters that lane only after the reader's record is sealed. The D_0 lane judges the original citations alone. The reconciler keeps a shared label or reports underdetermined. Any transport or parse failure is an execution status, never a scientific verdict.

Unaddressed
No reviewed relevant passage answers the claim under the stated corpus and retrieval budget.

Underdetermined
Relevant evidence exists, but disagreement, imprecision, or an unsupported inference prevents resolution.

SciFact's native label space is three-way, so the scored mapping is supported → SUPPORT, contradicted → REFUTE, and unaddressed or underdetermined → NEI. The four-way distinction is reported but not scored against gold.

3.3 Research questions

- **RQ1 (primary).** Does independent retrieval plus staged evidence-first judgment reduce false endorsement relative to same-evidence checking? (Specified before the held-out run; not publicly registered. The comparison is not compute-matched.)
- **RQ2 (descriptive).** How often do the two scopes disagree, and how often does the system defer?
- **RQ3 (deferred).** Do evidence-linked explanations help non-experts? No participant was recruited; Appendix D records the unused protocol.

4 System

Figure 1 shows the two evidence lanes. Roles are isolated functions with typed inputs; several roles may call the same model in separate stateless requests. Separate roles do not imply statistically independent errors.

Independent gatherer.
Given the neutral question, retrieves from the frozen abstract corpus. Three query forms are issued; the top candidates are merged and at most eight passages are kept.

Evidence reader.
Answers the neutral question from the shuffled passages and cites the identifiers it used. Its record is sealed before the claim is revealed.

Judge.
Sees the claim, the sealed record, and the passages, and assigns one of the four labels.

Citation auditor.
Judges the claim against D_0 alone. It never sees D_1.

Reconciler.
If the two four-way labels agree, that label is kept; otherwise the verdict is underdetermined. A claim can be unsupported by its own citation and supported elsewhere; the two facts are reported separately and are not repaired into one score.

The blinding is partial: a neutral question still reveals the topic. The isolation guarantee is structural (payload shape plus tests), not semantic. The evaluation scores crowd-sourced SciFact claims, not claims proposed from a generated report, so the proposer and the renderer are described in the repository and are not part of the scored comparison.

The system fails closed. A malformed response, a cited identifier that was not supplied, or an endpoint failure is recorded as `failed` or `timeout` and excluded from scored metrics. Failed rows are never remapped into a scientific verdict in the primary accounting. A stricter all-rows accounting that scores those rows as the conservative NEI is reported as sensitivity. The primary development and held-out runs do not retry a failed claim. A later development re-run retries up to four times; it is a second run, not a replacement (Section 6.9).

5 Experimental Setup

5.1 Data and splits

We use SciFact [Wadden et al., 2020]. The corpus is the released set of biomedical abstracts, frozen by hash. Official test labels are not used. Two samples of 300 claims are frozen by manifest: a development sample drawn with a fixed seed from the training pool (`dev300_seed42.json`), used for prompt decisions, and SciFact's public development split, held out locally and used once under the freeze record. Both samples group claims by cited-document family so that paraphrases do not straddle the comparison. The scored unit is a claim-level three-way label, not the official abstract-level or sentence-level SciFact decision.

SciFact has been public since 2020 and is a plausible pretraining contaminant [Jacovi et al., 2023]. Condition B0 (claim only, no evidence) is the only check on parametric leakage; we do not run a membership test.

5.2 Conditions

All conditions share the frozen corpus, the claim sample, the scoring code, and the prompt files. They differ in the intervention named in Table 1.

B2 is the evidence-first control specified for RQ1. It is not matched to V on inputs: B2's compare step sees the claim and the sealed reader record and emits a three-way label; V's judge also sees the passages and emits a four-way label. It is not matched on compute: V issues three calls per claim against B2's two, at about 6.4 times the tokens. A planned self-consistency control at matched tokens was not run.

5.3 Metrics and statistics

Verification quality is measured by per-class precision, recall and F1, plus micro and macro F1, on the native three-way label space. The primary metric is false endorsement: false SUPPORT predictions divided by the number of gold non-support claims. Coverage is the fraction of submitted claims that produced a scored prediction; fail-closed rows are counted.

Table 1: The condition matrix. "Adaptation" means a prompt-level implementation of the named mechanism, never a reproduction of a trained system. V-D1 is recovered from the V predictions by a mapping step, so it inherits V's execution failures.

ID	Condition	Evidence and reading policy
B0	Prior-knowledge judge	No evidence; claim only
B1	Atomic claim, claim-visible judge	D_0; claim-visible
B2	Neutral question, isolated reader, compare	D_0; evidence-first
B3	Fresh retrieval, direct judge	D_1; claim-visible
V-D1	Fresh retrieval, isolated reader, judge	D_1; evidence-first
V	Full staged validator	$D_0 \cup D_1$; reconciled
O	Evidence oracle (diagnostic)	Gold rationale sentences

The specified primary comparison is V versus B2 on false endorsement. The interval is a paired cluster bootstrap over cited-document families, 2000 resamples, 95% percentile interval; a claim enters only when both methods produced an execution-ok, gold-labelled row. Because V's false endorsement is zero, that interval's upper bound is pinned at zero. We therefore also report the exact two-sided McNemar test on the discordant pairs (B2 endorses and V does not, versus the reverse). With a baseline rate of about 1%, an exact McNemar test needs at least 6 one-sided discordant events to reach $p < 0.05$; the study was not powered for that. Secondary comparisons are exploratory. A Holm adjustment was computed on a family of one and is not reported.

5.4 Models

GPT-6 Luna is a hosted instruction-tuned model accessed through an OpenAI-compatible chat endpoint; it is the model behind the primary comparison, the held-out run, and the cost accounting. MiniMax M3.1 Flash is a second, architecturally unrelated model accessed through a local agent runtime exposed as a loopback chat endpoint, one fresh session per request. A fixed agent preamble is prepended to every MiniMax request, so its token counts are not comparable with Luna's and we do not report them. Temperature 0 is requested where the endpoint accepts it; the hosted OpenAI-compatible path used for Luna omits the field. Prompts are versioned text files under `prompts/`; their hashes enter the freeze record.

A thirty-claim subset of the development sample (ten SUPPORT, ten REFUTE, ten NEI, seed 42, selected before any sweep model was run) was reused with four closed-weight models from the same vendor lineage, including Luna. Open-weight models were attempted through free endpoints; none completed both conditions, so they contribute no row.

5.5 Protocol

The freeze record (`docs/freeze.json`, committed 2026-09-26) locks the model identifier, the corpus hash, and per-method configuration and prompt hashes. It was written from completed development sidecars after the development outcome was known, and it is not a public preregistration. The held-out run (2026-09-25) uses that frozen configuration and was not retried. The primary development numbers in this paper are the 2026-09-25 run under the same configuration. On 2026-10-20 the development sample was re-run after a retry policy was added; that re-run has a different configuration hash, recovered V's 32 failed claims, and flipped 26 of 300 B2 labels (9%). We report it as a second run (Section 6.9).

Table 2: Development sample ($n = 300$ claims). FE is false endorsement on the gold non-support denominator. Fail-closed rows are excluded from the scored metrics and accounted for in Table 12. Luna rows are the frozen-configuration run; MiniMax is a later development run of a second model.

Method	n	micro-F1	macro-F1	FE	SUPPORT R
GPT-6 Luna					
B0	300	0.557	0.517	0.448	0.872
B1	300	0.840	0.838	0.049	0.838
B2	300	0.747	0.755	0.011	0.530
B3	300	0.790	0.787	0.055	0.795
V-D1	268	0.743	0.745	0.019	0.598
V	268	0.675	0.673	0.000	0.383
O	300	0.857	0.860	0.016	0.735
MiniMax M3.1 Flash					
B1	300	0.863	0.856	0.093	0.966
B2	300	0.870	0.867	0.033	0.855
V-D1	300	0.853	0.851	0.033	0.829
V	300	0.830	0.837	0.016	0.692

Table 3: Held-out sample (SciFact public development split, used once). V-D1 is derived from V and inherits its 38 fail-closed rows.

Method	n	micro-F1	macro-F1	FE	SUPPORT R
GPT-6 Luna					
B2	300	0.760	0.765	0.017	0.581
V-D1	262	0.721	0.715	0.032	0.585
V	262	0.618	0.599	0.000	0.368

6 Results

6.1 Primary comparison

Table 2 gives the development results, Table 3 the held-out results, and Table 4 the specified paired comparison with an exact McNemar test. On the frozen Luna development sample, B2 reaches micro-F1 0.747 with false endorsement 0.011, while V reaches micro-F1 0.675 with false endorsement 0.000 on 268 scored claims (32 fail-closed). The paired difference is -0.012 ([-0.031, 0.000]) over 268 paired claims; the interval includes zero. The same comparison on 161 gold non-support claims is 2 versus 0 discordant pairs (exact McNemar $p = 0.50$). The held-out sample is 2 versus 0 ($p = 0.50$) with interval -0.013 ([-0.033, 0.000]). MiniMax is 5 versus 2 ($p = 0.45$). In every row the data are compatible with both no reduction and a reduction of a few points. That is an underpowered comparison, not a demonstration of no effect.

Figure 2 places each completed development condition by false endorsement and SUPPORT recall. V sits at zero false endorsement and the lowest support recall. B1, not B2, is the highest-F1 same-evidence cell.

The independent-retrieval arm, recovered from V, does not reduce false endorsement relative to B2. On development, V-D1's rate is 0.019 against B2's 0.011. On the held-out sample it is 0.032 against 0.017: the independent arm endorses more falsely than the same-evidence baseline. V's zero false endorsement is therefore a property of the reconciliation rule, not of retrieval.

6.2 The class trade-off

Table 5 breaks the two primary conditions down by class. V raises SUPPORT precision (from 0.969 to 1.000 on Luna development) by pushing uncertain claims into NEI, and it pays for that in SUPPORT recall (0.530 to 0.383; 55 missed supports under B2, 66 under V) and in REFUTE recall (0.790 to 0.627;

Table 4: V versus B2 on false endorsement. "B2 only" and "V only" are discordant pairs among gold non-support claims. McNemar p is the exact two-sided test. The percentile bootstrap interval is shown for continuity with the specified analysis; V's zero FE pins its upper bound at zero.

Model	Run	n	non-supp.	B2 only	V only	McNemar p	ΔFE [95% CI]
GPT-6 Luna	development	268	161	2	0	0.50	-0.012 [-0.031, 0.000]
GPT-6 Luna	development, second run	300	183	2	0	0.50	-0.011 [-0.029, 0.000]
GPT-6 Luna	held-out	262	156	2	0	0.50	-0.013 [-0.033, 0.000]
MiniMax M3.1 Flash	development	300	183	5	2	0.45	-0.016 [-0.046, 0.011]

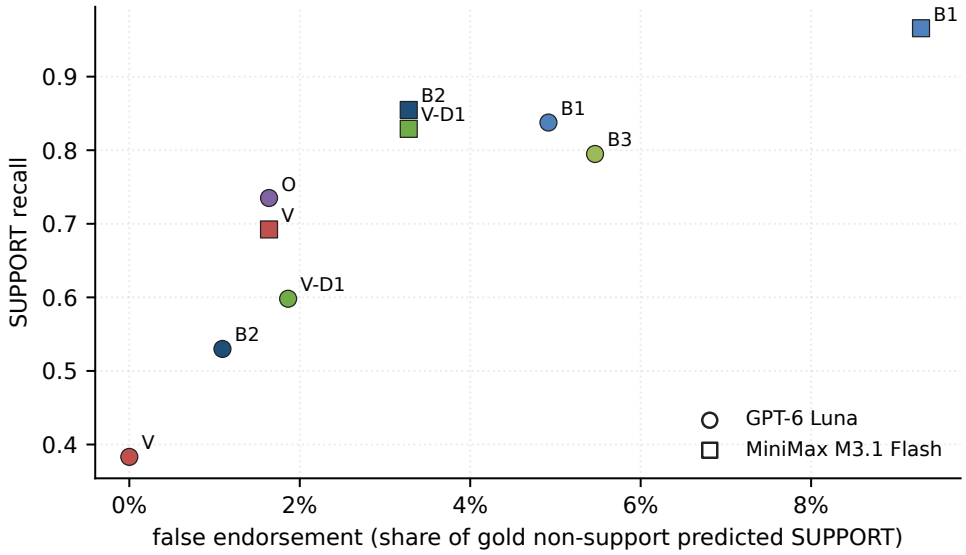


Figure 2: False endorsement against SUPPORT recall on the development sample. B0 (prior knowledge only) is omitted: its false endorsement (0.448) is an order of magnitude larger. Circles are Luna; squares are MiniMax.

13 missed contradictions under B2, 19 under V). MiniMax shows the same shape at a higher absolute level (0.855 to 0.692; 17 missed supports under B2, 36 under V). The cost on the full sample is missed support first and missed contradiction second. Figure 3 makes the Luna counts visible.

6.3 The 2×2 cells

Table 6 separates evidence access from reading policy as far as the condition matrix allows. The cells are not a clean factorial: they differ in what the final step sees and in the label space, and V-D1 is derived from V. Read as a descriptive grid, two patterns appear under Luna. Claim-visible judging beats evidence-first judging on micro-F1 (B1 0.840 against B2 0.747; B3 0.790 against V-D1 0.743). Evidence access barely moves the evidence-first cell (V-D1 FE 0.019 against B2 0.011). MiniMax does not repeat the claim-visible advantage (B1 0.863 against B2 0.870) and did not complete B0, B3, or O. We therefore treat the grid as a Luna-development description, not as a replicated factorial.

6.4 Where the abstentions come from

V stores its four-way label alongside the mapped native label. On the frozen development sample it predicts NEI on 225 of 300 claims, of which 60 are unaddressed and 165 are underdetermined (Figure 4). SciFact's gold cannot score that split.

The more informative split is whether the two scopes agreed. Table 7 conditions accuracy on that split, now separated into three groups: the same four-way label; both no-verdict labels (unaddressed versus underdetermined, both NEI); and a genuine disagreement on the scored three-way label. When

Table 5: Per-class precision, recall, and F1 on the development sample, with confusion counts. Luna is the frozen-configuration run.

Method	class	P	R	F1	TP	FP	FN
GPT-6 Luna							
B2	SUPPORT	0.969	0.530	0.685	62	2	55
B2	REFUTE	0.875	0.790	0.831	49	7	13
B2	NEI	0.628	0.934	0.751	113	67	8
V	SUPPORT	1.000	0.383	0.554	41	0	66
V	REFUTE	0.941	0.627	0.753	32	2	19
V	NEI	0.560	0.982	0.713	108	85	2
MiniMax M3.1 Flash							
B2	SUPPORT	0.943	0.855	0.897	100	6	17
B2	REFUTE	0.800	0.903	0.848	56	14	6
B2	NEI	0.847	0.868	0.857	105	19	16
V	SUPPORT	0.964	0.692	0.806	81	3	36
V	REFUTE	0.944	0.823	0.879	51	3	11
V	NEI	0.722	0.967	0.827	117	45	4

Table 6: Condition grid on the development sample. V-D1 is derived from the V predictions, not re-run.

Cell	Evidence	Reading policy	micro-F1	FE
GPT-6 Luna				
B0	none	prior knowledge	0.557	0.448
B1	D0	claim-visible	0.840	0.049
B2	D0	neutral / evidence-first	0.747	0.011
B3	D1	claim-visible	0.790	0.055
V-D1	D1	neutral / evidence-first	0.743	0.019
V	D0 ∪ D1	reconciled	0.675	0.000
O	gold rationales	claim-visible	0.857	0.016
MiniMax M3.1 Flash				
B1	D0	claim-visible	0.863	0.093
B2	D0	neutral / evidence-first	0.870	0.033
V-D1	D1	neutral / evidence-first	0.853	0.033
V	D0 ∪ D1	reconciled	0.830	0.016

the four-way labels agree, V matches B2 (0.828 against 0.828 on 163 development claims). When both scopes already say NEI in different words (50 claims), V abstains and is right about as often as B2 (0.580 against 0.600). When the scored labels differ (55 claims), V abstains on every claim by construction and is right on 0.309 of them; B2 is right on 0.618. The held-out sample is sharper: B2 is right on 0.700 of the 50 three-way conflicts, V on 0.100. "Disagreement drives abstention" is the algorithm. The empirical content is that the conflict signal is a weak predictor of a B2 error.

6.5 Alternative decision rules

Because V stores both scope labels, alternative reconciliation rules can be applied without a new model call (Table 8). Trusting D_1 alone recovers B2's development accuracy (0.743) at a higher false-endorsement rate (0.019). Trusting D_0, or falling back from D_0 to D_1 when D_0 has no verdict, raises accuracy and false endorsement together. No saved-label rule is both more accurate than B2 and lower in false endorsement. On the held-out sample, trusting D_1 alone is the cell that produces the 0.032 false-endorsement rate already noted.

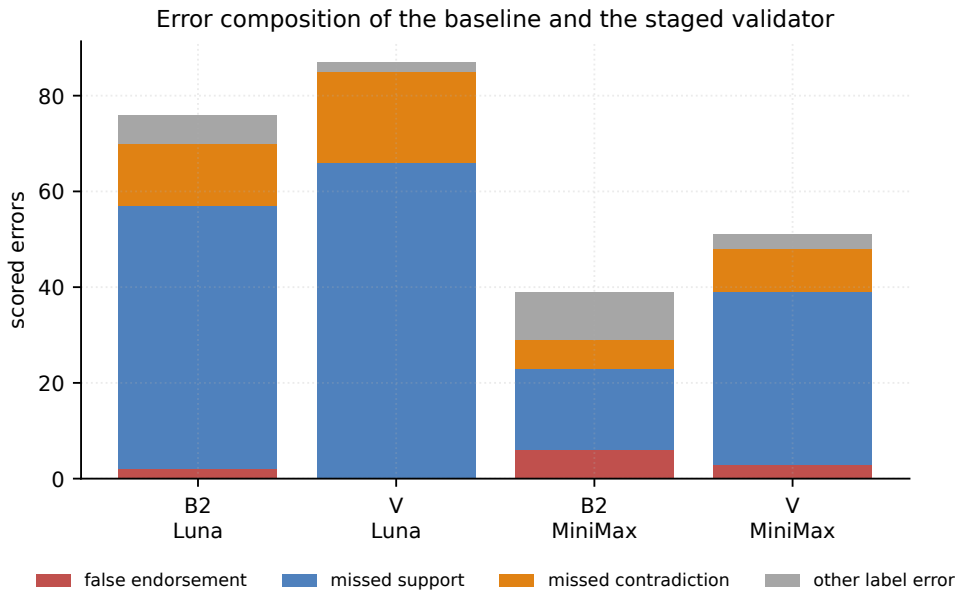


Figure 3: Scored error counts for B2 and V on the frozen development sample. Execution failures are excluded and counted in Table 12.

6.6 Evidence acquisition

Table 9 reports what independent retrieval bought. Recall@k is 0.85 over the 179 claims that have annotated evidence, but every annotated document it finds was also cited: unique yield is 0.00. In SciFact the cited document set contains the annotated document set, so a retriever that searches the same corpus cannot find new gold evidence. It can only add unannotated context, and 93% of what it returns is unjudged. The independent arm therefore cannot repair a wrong citation on this corpus, because a wrong citation is, by construction, a document without a rationale. That property bounds every claim in this paper. The experiment that would test the motivating hypothesis—swap D_0 for non-gold abstracts and ask whether D_1 recovers the gold rationale—was not run.

6.7 A second model and a thirty-claim subset

MiniMax M3.1 Flash, reached through a different harness, reproduces the direction of the V-versus-B2 contrast on the development sample (Table 4): V is more conservative, SUPPORT recall falls, and the McNemar test does not reject. MiniMax is a markedly better verifier in absolute terms (B2 micro-F1 0.870 against 0.747). The two models are not matched for size, training, or harness, and MiniMax has no held-out run and an incomplete condition matrix (B0, B3, O, and calibration never completed: the free endpoint that served it was withdrawn). It corroborates the specified contrast, not the wider grid.

To check whether the conservative shift belongs only to these two models we repeated B2 versus V on a frozen thirty-claim subset with four closed-weight models from one vendor lineage, including Luna (Table 10, Figure 5). Thirty claims can resolve a difference of several claims, not a one-point false-endorsement rate; every ΔFE interval collapses to $[0, 0]$. What the subset can show is where a shift lands: REFUTE recall falls under V in all four models, while SUPPORT recall is unchanged in three of the four. That concentration does not match the full-sample class breakdown (Table 5), so the subset is a directional check on a handful of claims, not a finding about the cost of staging.

6.8 Cost

Table 11 reports model calls and measured tokens for Luna, now taken from the same development run for every condition. V issues three calls per claim against B2's two. A B2 call measures about 605 tokens

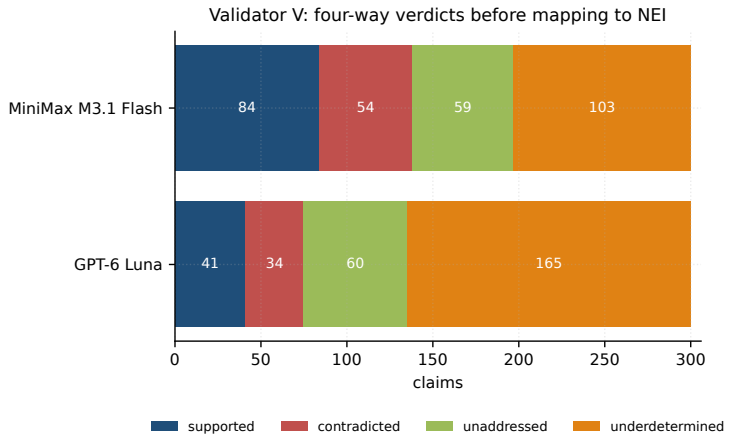


Figure 4: V's four-way verdicts on the frozen development sample, before the mapping onto SciFact's three-way label space.

Table 7: Accuracy conditioned on the two evidence scopes. "Both no verdict" means the four-way labels differ but both map to NEI. "Different scored label" means SUPPORT, REFUTE, and NEI disagree. V abstains on every row of the last two groups by construction. Fail-closed rows are excluded.

Scopes	n	B2 acc.	V acc.	V NEI
GPT-6 Luna, development				
same four-way label	163	0.828	0.828	0.540
both no verdict, different kind	50	0.600	0.580	1.000
different scored label	55	0.618	0.309	1.000
GPT-6 Luna, held-out				
same four-way label	139	0.871	0.842	0.496
both no verdict, different kind	73	0.603	0.548	1.000
different scored label	50	0.700	0.100	1.000
MiniMax M3.1 Flash, development				
same four-way label	214	0.911	0.925	0.355
both no verdict, different kind	36	0.861	0.917	1.000
different scored label	50	0.700	0.360	1.000

and a V call about 2,579, so V costs about 6.4 times B2 per claim. In exchange V produces two scoped judgments instead of one. A deployment that only needs a single verdict is better served by B2; the value of V is that its two scopes are separately auditable. MiniMax has no cost row: its harness preamble dominates every prompt.

6.9 Fail-closed rows and the second development run

Table 12 repeats B2 and V under both accountings. On the frozen development run, scoring V's 32 fail-closed rows as NEI lowers micro-F1 from 0.675 to 0.640; false endorsement stays at zero. Those 32 claims are not a random slice: gold is SUPPORT on 10, REFUTE on 11, and NEI on 11. The held-out run, which also does not retry, lost 38 of 300 claims, of which 18 are gold SUPPORT; all-rows micro-F1 falls from 0.618 to 0.583. The runner overwrites the failure reason at reconcile time, so timeout, parse, and transport cannot be separated.

The 2026-10-20 development re-run recovered every failed V request and left V's false endorsement at zero (SUPPORT recall 0.427). It is not interchangeable with the frozen run: B2 flipped 26 of 300 labels (9%) and V flipped 14 of 268 previously-ok labels (5%) at temperature 0. The specified contrast (V more conservative, interval covering zero, McNemar $p = 0.50$ on 2 versus 0 pairs) is the same in both runs.

Table 8: Alternative decision rules applied to the saved per-scope labels, on claims where both B2 and V completed. No model is called.

Decision rule	accuracy	FE	SUPPORT R	REFUTE R
GPT-6 Luna, development ($n = 268$)				
B2 baseline (reference)	0.743	0.012	0.523	0.804
Both scopes must agree (V as run)	0.675	0.000	0.383	0.627
Citation scope only (D0)	0.683	0.019	0.430	0.686
Independent scope only (D1)	0.743	0.019	0.598	0.745
D0, falling back to D1 when D0 has no verdict	0.750	0.037	0.645	0.804
GPT-6 Luna, held-out ($n = 262$)				
B2 baseline (reference)	0.763	0.013	0.585	0.754
Both scopes must agree (V as run)	0.618	0.000	0.368	0.456
Citation scope only (D0)	0.653	0.000	0.415	0.526
Independent scope only (D1)	0.721	0.032	0.585	0.614
D0, falling back to D1 when D0 has no verdict	0.756	0.032	0.632	0.684
MiniMax M3.1 Flash, development ($n = 300$)				
B2 baseline (reference)	0.870	0.033	0.855	0.903
Both scopes must agree (V as run)	0.830	0.016	0.692	0.823
Citation scope only (D0)	0.847	0.038	0.769	0.919
Independent scope only (D1)	0.853	0.033	0.829	0.823
D0, falling back to D1 when D0 has no verdict	0.873	0.055	0.906	0.919

Table 9: Evidence acquisition for the independent-retrieval arm on the development sample.

Quantity	Value
Claims scored	300
Claims with annotated evidence	179
Passages retrieved per claim (k)	8
Recall@k (annotated documents found)	0.849
Precision@k (retrieved documents annotated)	0.074
Overlap with D_0 (retrieved documents also cited)	0.096
Unique yield (annotated found but not cited)	0.000
Refuting yield (contradicting documents found)	0.171
Unjudged rate (no annotation)	0.926
Retrieved documents	2400
Annotated documents retrieved that were not cited	0

7 Discussion

The staged validator did not significantly reduce false endorsement relative to a baseline that already read the original citations. The correct reading of that sentence is narrow. The comparison is underpowered (two discordant events on Luna, five versus two on MiniMax). The comparison is not compute-matched. The comparison is not a test of independent retrieval as a repair for bad citations, because SciFact never presents a bad citation. What the data do show is that adding a second reader of the same literature, and abstaining when the two readers disagree, produces a conservative operating point whose errors are mostly missed support.

Staging is not free of side effects even as a conservative policy. On the claims where the scored labels differ, B2 is more often right than the gold-NEI rate that V achieves by abstaining. A reader who treats every abstention as a signal that the claim is unsupported would be making a new error; that is why the interface keeps the scientific verdict ("underdetermined") separate from the system action ("defer").

A better decision procedure for the conflict case is the obvious next step: trust one scope, require three-way rather than four-way agreement, or hand the conflict to a reader. Table 8 shows that none of the

Table 10: B2 versus V on the frozen thirty-claim subset. Cells after the interval are "B2 / V". The subset is descriptive: read it for direction, not for significance. All four models are closed-weight models from one vendor lineage; Luna is the primary model, not a held-out replicate.

Model	Weights	ΔFE [95% CI]	SUPPORT R	REFUTE R	micro-F1
GPT-6 Astra	closed	+0.000 [0.000, 0.000]	0.700 / 0.600	0.900 / 0.500	0.833 / 0.700
GPT-6 Sol	closed	+0.000 [0.000, 0.000]	0.700 / 0.700	0.800 / 0.500	0.833 / 0.733
GPT-5.6 Terra	closed	+0.000 [0.000, 0.000]	0.700 / 0.700	0.900 / 0.500	0.867 / 0.733
GPT-6 Luna	closed	+0.000 [0.000, 0.000]	0.500 / 0.500	0.900 / 0.500	0.800 / 0.667

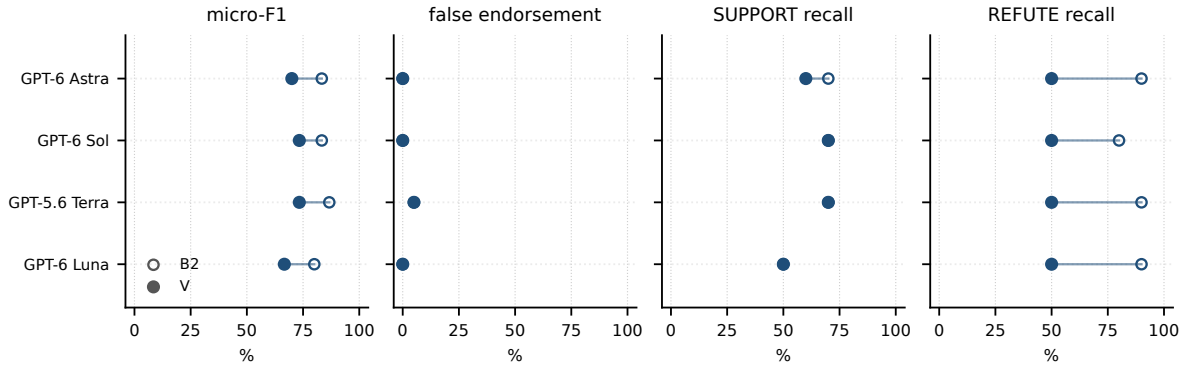


Figure 5: B2 to V on the thirty-claim subset. Hollow marks are B2, filled marks are V. The largest false-endorsement rate in any row is 5%.

saved-label shortcuts is free. The experiment that would reopen RQ1 is a citation-corruption condition on this corpus, or a move to SciFact-Open, where independent retrieval can actually find new evidence.

8 Limitations

- **The benchmark cannot test the motivating mechanism.** Unique yield is zero by construction. A citation-swap experiment was not run.
- **The primary test is underpowered** and the comparison is not compute-matched. RQ1's "matched inference budget" was not implemented.
- **No public preregistration.** The freeze record is an internal hash lock written after the development outcome was known. The primary development numbers use that frozen configuration; a later retry re-run is reported separately.
- **The 2×2 is confounded** by final-step inputs, label space, and a derived V-D1 arm.
- **Four-way labels are not human-validated.** The unaddressed / underdetermined split is descriptive.
- **One corpus, one domain, abstract-level text.** English biomedical abstracts. Claims that need methods, figures, or supplements are out of scope.
- **Supplied claims, not generated reports.** The proposer is not scored.
- **The two models are not matched**, MiniMax's matrix is incomplete, and the thirty-claim subset is four closed-weight models from one vendor lineage, including the primary model.
- **SciFact contamination is unchecked** beyond B0 (SUPPORT recall 0.872 from the claim alone).
- **No user study.** See RQ3.

Table 11: Calls per claim and measured tokens for GPT-6 Luna, taken from the development run that recorded the most calls for each condition. Estimated tokens per claim is the product of the two preceding columns. Resumed runs keep only the last invocation's usage record, so this is an estimate, not a per-claim total.

Method	measured on	calls/claim	tokens/call	est. tokens/claim
GPT-6 Luna				
B1	dev	1	752	752
B2	dev	2	605	1,209
B3	dev	1	3,072	3,072
V	dev	3	2,579	7,737
O	dev	1	380	380

Table 12: Sensitivity to the fail-closed accounting and to the second development run. "Scored only" excludes execution failures; "all rows" scores every submitted claim with the label the method recorded.

Method	accounting	n	micro-F1	FE	SUPPORT R
GPT-6 Luna					
B2 (dev300)	scored only	300	0.747	0.011	0.530
B2 (dev300)	all rows	300	0.747	0.011	0.530
V (dev300)	scored only	268	0.675	0.000	0.383
V (dev300)	all rows	300	0.640	0.000	0.350
B2 (dev300rerun)	scored only	300	0.747	0.011	0.573
B2 (dev300rerun)	all rows	300	0.747	0.011	0.573
V (dev300rerun)	scored only	300	0.687	0.000	0.427
V (dev300rerun)	all rows	300	0.687	0.000	0.427
B2 (heldout300)	scored only	300	0.760	0.017	0.581
B2 (heldout300)	all rows	300	0.760	0.017	0.581
V (heldout300)	scored only	262	0.618	0.000	0.368
V (heldout300)	all rows	300	0.583	0.000	0.315
MiniMax M3.1 Flash					
B2 (dev300)	scored only	300	0.870	0.033	0.855
B2 (dev300)	all rows	300	0.870	0.033	0.855
V (dev300)	scored only	300	0.830	0.016	0.692
V (dev300)	all rows	300	0.830	0.016	0.692

9 Ethics

The task is interpreting published abstracts, not giving medical advice. The rendered output is a label, a rationale, and a search scope; it is not a clinical recommendation. Abstention has its own risk: a reader who sees an abstention may infer that the claim is false. Any deployment should display the scientific verdict and the system action separately. The MiniMax harness is an interactive coding runtime; we ran it with a fixed preamble, a fixed corpus, and no tool-enabled tasks.

10 Conclusion

A staged, evidence-first validator for scientific claims—a sealed reader, a claim-visible judge, a citation auditor, and a reconciler that reports two evidence scopes separately—did not significantly reduce false endorsement relative to a same-evidence baseline on SciFact. It became a conservative checker whose errors come mostly from disagreement between two readers of the same literature, and whose conservatism is a reconciliation rule rather than a retrieval gain. Independent retrieval cannot find new gold evidence on this benchmark. We report that as an inconclusive result, with the mechanism, the cost, the second run, and the artifact needed to re-run it. The next step is a decision procedure for scope disagreement, and an experiment in which the original citations can actually be wrong.

A Prompts and output contracts

The prompts are versioned files under `prompts/`; the run sidecar records each file's SHA-256 and the bundle hash. The contracts are:

Reader (claim-blind). Input: a neutral question and a list of passages with identifiers. Output: `{"answer": str, "cited_doc_ids": [int]}`. The reader is not shown the claim, the asserted answer, the report, or any gold rationale.

Judge (claim-visible). Input: the atomic claim, the sealed reader record, and the passages. Output: `{"label": one of supported, contradicted, unaddressed, underdetermined; "rationale_codes": [str]; "cited_span_ids": [str]; "uncertainty_reasons": [str]}`.

Citation auditor (D0 only). Input: the atomic claim and the original cited passages. Output: the judge contract plus `{"citations": [{"span_id": str, "adequacy": adequate | inadequate | unverifiable, "defects": [str]}]}`.

Compare (B2). Input: the claim, the neutral question, and the sealed reader record. Output: `{"label": SUPPORT | REFUTE | NEI, "rationale": str}`.

The parsers reject an out-of-set label, a cited identifier that was not supplied, and a missing mandatory field.

B Metric definitions

Let S be the set of scored claims, $g_i \in \{\text{SUPPORT}, \text{REFUTE}, \text{NEI}\}$ the gold label, and p_i the predicted label. The endorsement class is $E = \text{SUPPORT}$.

$$\text{FE}_{\text{gold}} = \frac{|\{i : p_i = E,\ g_i \neq E\}|}{|\{i : g_i \neq E\}|} \qquad \text{FE}_{\text{pred}} = \frac{|\{i : p_i = E,\ g_i \neq E\}|}{|\{i : p_i = E\}|} \tag{1}$$

Micro-F1 pools the confusion matrix over the three classes; macro-F1 averages the three per-class F1 values. Coverage is the fraction of submitted claims that produced a scored prediction. The paired bootstrap resamples cited-document families with replacement (2000 resamples). The exact McNemar p-value on b and c discordant pairs is $2 \sum_{i=0}^{\min(b,c)} \binom{b+c}{i} 2^{-(b+c)}$, capped at 1.

C Case records

The three rows below are selected deterministically from the saved predictions: the first frozen-development claim whose scopes disagreed on the scored label, the first whose scopes agreed on underdetermined while B2 and gold said SUPPORT, and the first B2 false endorsement.

Table 13: Case studies from the frozen development sample.

Case and claim	Gold	B2	V (D0 / D1)	What the record shows
Scope conflict, abstained *Rapid phosphotransfer rates are correlated with histidine kinase regulator specificity.*	SUPPORT	NEI	NEI (underdetermined / supported)	The independent scope finds support the citation scope does not; the reconciler refuses to choose and abstains.
Baseline false endorsement *Beta-band coherence is enhanced for visible stimuli over invisible stimuli.*	NEI	SUPPORT	NEI (supported / unaddressed)	The baseline endorses from the original citations; the validator abstains, so the endorsement is caught but not replaced.
Coverage loss, scopes agree *ALDH1 expression is associated with poorer prognosis for breast cancer primary tumors.*	SUPPORT	SUPPORT	NEI (underdetermined / underdetermined)	Both scopes call the evidence inconclusive; the validator abstains on a claim the baseline and the gold endorse.

D Protocol for the deferred human study

No participant was recruited and no ethics review was obtained. The protocol is recorded so the absence of results is explicit.

Design. Thirty non-expert adults (a feasible range of 24–36), plus four to six pilot participants, would each see twelve short, expert-reviewed items. Condition assignment is counterbalanced: (A) conclusion plus original citations; (B) A plus the validator verdict; (C) B plus the evidence-linked explanation. Conditions B and C must use the same system verdict.

Outcomes. Pre- and post-interface 0–100 judgements that the conclusion is adequately supported, a four-way evidence-status choice, one comprehension question, a use-or-defer decision, and response time. The primary exploratory outcome is the post-interface Brier score for evidence support comparing C with B.

E Reproduction

1. Install the package and its development extras, and verify the pinned corpus hash (`data/pins/scifact/`).
2. Start the model endpoint. For Luna, set `OPENAI_API_KEY` and `OPENAI_MODEL`. For MiniMax, start a local agent server and the loopback bridge in `tools/opencode_bridge.py`.
3. The frozen-configuration development run is the artifact under `artifacts/validator_v_dev300` and `artifacts/same_evidence_b2_dev300`. The 2026-10-20 re-run is under `artifacts/api/dev300/`. The held-out run is under `artifacts/held_out/` and requires `--frozen-final`.
4. Regenerate every table and figure (`scripts/paper/make_tables.py`, `scripts/paper/make_figures.py`) and rebuild the PDF (`scripts/paper/build_pdf.sh`).

Every number in this paper is read from the generated LaTeX fragments under `paper/tables/`. The prose macros in `paper/tables/macros.tex` are generated by the same script.

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